

I N D E X

NAME: Kuberan STD.: AI SEC.: A ROLL NO.: _____ SUB: Deep learning

S. No.	Date	Title	Page No.	Teacher's Sign / Remarks
1	24/7	Exploring Deep learning Platform		11/11/25
2	31/07	Implement a classifier using open source dataset		11/11/25
3	14/08	study of the classifiers with respect to statistical parameters		11/11/25
4	14/08	Build a simple feed forward neural network to register uc		11/11/25
5	22/08	study of Activation function and its role		11/11/25
6	09/09	Gradient descent & backpropagation in deep neural networks		11/11/25
7	16/09	Build a cnn model classify cat and dog image		11/11/25
8	9/10	experiment of LSTM		} 11/11/25
9	09/10	Build a Recurrent Neural Network		
10	17/10	Perform Compression on mnist dataset using Autoencoder		} 11/11/25
11	17/10	Experiments using variational Autoencoder (VAE)		
12	26/10	Implement a Deep convolutional GAN (DCGAN) to color image		} 11/11/25
13	26/10	Understand the Architecture of a pre-trained Model		
14	26/10	Implement a pre-trained CNN model as a Feature Extractor		} 11/11/25
15	26/10	Implement a yolo model for object Detection		

26/10/25

Ex: NO: 13

13. Understanding The Architecture of a Pre-trained Model

Aim:

To study and analyze the internal architecture of a pre-trained CNN model (e.g. VGG16, ResNet, or Inception).

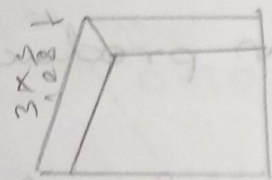
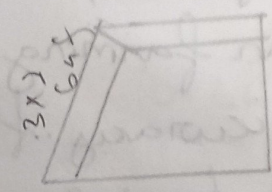
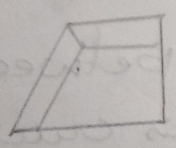
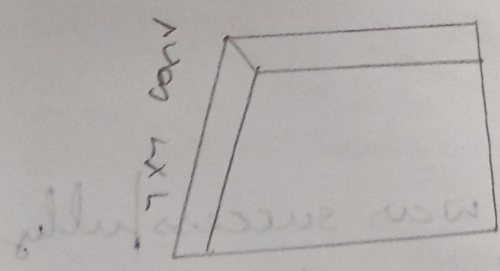
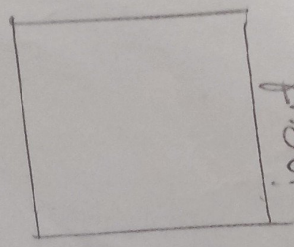
Objectives

1. To understand how deep pre-trained networks are constructed
2. To analyze convolutional layers, pooling layers, and fully-connected layers
3. To observe how pre-trained weights encode hierarchical image features

Pseudocode

1. Start
2. Import a pre-trained CNN model (e.g. VGG16) with weights = 'imagenet'.
2. Display model summary.
3. Visualize selected intermediate layers and their feature maps
4. Interpret how features transition from edges - features - objectives

input
image



3
2
1

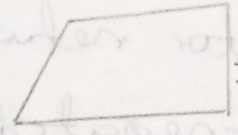
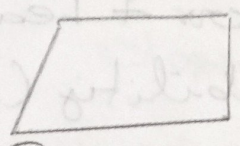
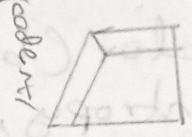
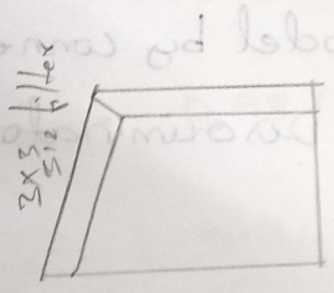
out

1

1. Generator begins generating plausible
patterns initially
2. The generator produces a noisy shape
and features begin forming
3. The generator produces a cleaner shape
and features begin forming
4. The generator produces a more refined shape
and features begin forming

Observation:

End



two multiplications
output

A. Compile Discriminator (GanNet)
GanNet (GanNet, Adam)

5. Create GAN model by connecting
Generator → Discriminator

Epoch
Output

Layer

Feature type

Description

Conv 1

Edges, colors

Basic visual features

Layer 1

Features

Detect stripes

Layer 2

Shapes

Recognize patterns

Layer 3

Object

Detect object

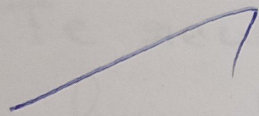
Layer 4

Semantic

High level
context

observation

1. Early layers detected simple shapes like edges and color gradients
2. Middle layers captured textures and repeated patterns
3. Visualize selected intermediate layers and their feature semantic representation
4. Pre-trained models reduce training time and improve accuracy significantly



Result

eg. 1 The internal working of a pre-trained CNN model was successfully studied showing how deep networks learn hierarchical visual features.